

NESBYEN-TODOKK, Norway

WMO: 013730

Lat: 60.567N

Lon: 9.133E

Elev: 166

StdP: 99.35

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-20.5	-25.2	0.4	-22.8	-22.5	0.5	-20.3	5.5	2.8	4.5	2.5	0.2	0	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	27.5	16.0	25.3	15.1	23.1	14.4	17.5	23.3	16.6	22.1	15.7	20.9	2.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.3	18.5	14.7	10.6	18.0	13.7	10.0	17.0	49.6	23.2	47.0	21.8	44.5	21.0	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.5	3.8	DB	-24.0	30.1	3.3	2.1	-26.4	31.5	-28.3	32.7	-30.2	33.9	-32.6	35.4	(4)
(5)			WB	-24.2	18.9	3.1	1.2	-26.4	19.8	-28.2	20.5	-30.0	21.1	-32.3	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-8.2	-6.0	-0.5	4.8	9.6	14.0	16.7	14.4	10.3	3.7	-2.3	-6.9	(6)	
	DBStd	9.61	6.40	5.98	4.77	3.08	3.86	2.90	2.85	2.43	3.04	4.02	4.73	6.75	(7)	
	HDD10.0	2681	565	448	326	158	55	3	0	2	33	199	369	522	(8)	
	HDD18.3	5184	823	682	584	407	271	132	69	123	239	454	619	781	(9)	
	CDD10.0	561	0	0	1	2	43	124	206	138	43	3	0	0	(10)	
	CDD18.3	22	0	0	0	0	1	3	17	1	0	0	0	0	(11)	
	CDH23.3	422	0	0	0	0	32	119	246	23	1	0	0	0	(12)	
	CDH26.7	87	0	0	0	0	4	24	58	1	0	0	0	0	(13)	
(14) Wind	WSAvg	1.2	0.8	0.9	1.4	1.6	1.6	1.6	1.4	1.2	1.2	0.9	0.8	0.8	(14)	
(15) Precipitation	PrecAvg	557	37	23	22	27	43	66	83	79	53	50	43	31	(15)	
	PrecMax	740	64	51	57	60	136	142	186	167	120	139	154	56	(16)	
	PrecMin	390	9	2	0	3	0	4	20	16	18	6	8	4	(17)	
	PrecStd	74	14	13	13	17	27	33	38	35	21	31	27	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	9.6	14.6	19.7	26.9	29.5	30.5	26.0	22.4	17.7	11.4	9.2	(19)	
		MCWB	4.8	5.2	7.6	10.1	14.6	15.5	16.8	15.7	14.7	13.0	7.9	5.1	(20)	
	2%	DB	4.3	6.1	11.5	16.5	23.3	26.6	28.4	23.4	19.9	13.6	8.3	6.2	(21)	
		MCWB	1.5	2.6	5.8	8.6	12.8	14.8	16.9	15.9	13.3	9.3	6.0	3.1	(22)	
	5%	DB	2.0	3.8	9.1	14.3	20.9	24.2	26.3	21.6	18.0	11.9	5.8	4.2	(23)	
		MCWB	0.6	1.4	4.2	7.3	11.9	13.6	16.1	14.9	12.5	8.8	4.2	1.9	(24)	
	10%	DB	0.2	1.7	6.8	12.0	18.1	21.7	24.0	19.8	16.3	10.2	3.5	1.7	(25)	
		MCWB	-0.6	0.2	2.5	6.0	11.1	12.6	15.3	13.9	11.7	7.5	2.6	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	5.2	8.2	11.0	16.1	16.7	19.1	18.0	15.6	13.1	8.6	5.5	(27)	
		MCDB	8.3	9.1	14.7	16.8	23.3	23.0	25.5	22.6	19.7	16.9	10.7	8.7	(28)	
	2%	WB	2.1	2.9	6.2	9.2	14.3	15.8	18.0	16.8	14.4	10.5	6.4	3.5	(29)	
		MCDB	3.8	5.8	11.2	15.4	20.9	23.7	24.6	21.1	18.3	12.4	8.0	5.9	(30)	
	5%	WB	0.8	1.6	4.6	7.8	12.8	14.8	17.2	15.9	13.4	9.3	4.3	1.9	(31)	
		MCDB	1.9	3.5	8.8	13.1	18.8	21.3	23.3	19.8	16.6	11.5	5.5	3.8	(32)	
	10%	WB	-0.6	0.4	2.8	6.6	11.4	13.9	16.3	15.1	12.4	7.7	2.6	0.4	(33)	
		MCDB	0.3	1.8	6.3	11.4	16.8	19.7	21.9	18.6	15.2	9.9	3.5	1.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	8.0	10.8	12.0	12.7	12.2	11.5	10.3	10.6	7.7	5.4	6.3	(35)	
		MCDBR	9.0	10.3	13.5	16.7	17.6	18.1	16.3	13.2	13.5	10.4	8.1	10.1	(36)	
	5% WB	MCWBR	7.1	7.3	8.3	9.4	8.7	8.0	6.6	6.4	7.9	6.7	6.2	7.4	(37)	
		MCWBR	8.6	9.4	13.3	15.1	14.6	13.5	12.8	10.8	11.5	8.6	7.3	9.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.231	0.275	0.311	0.353	0.358	0.352	0.369	0.361	0.325	0.308	0.250	0.201	(40)	
		taud	2.156	2.248	2.227	2.240	2.390	2.450	2.418	2.432	2.517	2.466	2.236	2.051	(41)	
	Ebn at Noon		544	724	809	839	867	879	851	829	805	691	512	411	(42)	
		Edh at Noon	38	66	94	112	105	101	102	93	72	55	36	26	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.32	0.98	2.48	4.14	4.79	5.16	4.63	3.65	2.40	1.12	0.40	0.19	(44)	
	RadStd		0.06	0.21	0.20	0.55	0.53	0.49	0.58	0.32	0.31	0.20	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page